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Practice Assignment • 30 min

Due Jul 13, 11:59 PM IST

Your grade: 80%

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1. What is the probability of getting 20 heads in a row when flipping a fair coin? 1 / 1 point

- ○ $\frac{1}{2^{10}}$
- ○ $\frac{1}{10^{20}}$
- ◉ $\frac{1}{2^{20}}$
- ○ $\frac{1}{2^{40}}$

✓ Nice work

Correct. The probability of flipping a head 20 times in a row is $\frac{1}{2^{20}}$.

2. In the context of statistics, what is the arithmetic mean of the following set of numbers: 3, 3, 3, 7, 10, 5? 1 / 1 point

- ○ 4.8
- ◉ 5.167
- ○ 6
- ○ 5.5

✓ Nice work

Correct. The arithmetic mean is calculated as $\frac{(3+3+3+7+10+5)}{6}$.

3. What does the term 'mode' refer to in statistics? 1 / 1 point

- ◉ The most frequently occurring value in a data set
- ○ The middle value of a data set when arranged in order
- ○ The average value of a data set
- ○ The range of the data set

✓ Nice work

Correct. The mode is the value that appears most often in a data set.

4. Which measure of central tendency is most appropriate for skewed distributions? 1 / 1 point

- ○ Mean
- ○ Range
- ◉ Median
- ○ Mode

✓ Nice work

Correct. The median is less affected by outliers and provides a better central tendency measure for skewed distributions.

5. How do you calculate the standard deviation from the variance? 1 / 1 point

- ○ Subtract the variance from the mean
- ○ Divide the variance by the mean
- ◉ Take the square root of the variance
- ○ Square the variance

✓ Nice work

Correct. The standard deviation is the square root of the variance.

6. Given the following data set: 4, 8, 6, 5, 3, 7, 2, 9, calculate the arithmetic mean. 1 / 1 point

5.5

✓ Nice work

The arithmetic mean is 5.5.

1. Sum the values in the data set:

$4+8+6+5+3+7+2+9=44$

2. Divide the sum by the number of values:

$\frac{44}{8}=5.5$

7. Given the data set: 15, 18, 22, 20, 15, 18, 16, find the mode.

If the answer consists of multiple values, separate them with a comma, without spaces. 1 / 1 point

15,18

✓ Nice work

The mode is 15 and 18.

1. List the frequency of each value:

15: 2 times

18: 2 times

22: 1 time

20: 1 time

16: 1 time

2. Identify the most frequently occurring values:

15 and 18 each occur twice.

3. Therefore, the modes are 15 and 18.

8. Calculate the variance and standard deviation for the data set: 2, 4, 6, 8, 10.

Give the variance first, followed by the standard deviation, separated by a comma without spaces. 0 / 1 point

10,3.163

⊗ Try again

The variance is 8 and the standard deviation is 2.83.

1. Calculate the mean:

$\text{Mean}=\frac{2+4+6+8+10}{5}=6$

2. Subtract the mean from each data point and square the result:

$(2-6)^2=16$

$(4-6)^2=4$

$(6-6)^2=0$

$(8-6)^2=4$

$(10-6)^2=16$

3. Sum the squared differences:

$16+4+0+4+16=40$

4. Divide by the number of observations minus one (n-1) for sample variance:

$\frac{40}{4}=10$

5. The variance is 10.

6. Take the square root of the variance for the standard deviation:

$\sqrt{10}\approx 3.16$

9. A drawer contains six socks: one green, two blue and three red. What is the probability of drawing two blue socks in a row without replacement? 1 / 1 point

1/15

✓ Nice work

The probability is $\frac{1}{15}$.

1. Calculate the probability of drawing the first blue sock:

$P(\text{First blue})=\frac{2}{6}=\frac{1}{3}$

2. Calculate the probability of drawing the second blue sock:

$P(\text{Second blue} \mid \text{First blue})=\frac{1}{5}$

3. Multiply the probabilities:

$P(\text{Two blue socks})=\frac{1}{3}\times\frac{1}{5}=\frac{1}{15}$

10. If 30 people are in a room, what is the probability that at least two of them share the same birthday (assuming 365 days in a year)? 0 / 1 point

0.706

⊗ Try again

The probability is approximately 70.6%.

1. Calculate the probability that no two people share the same birthday:

$P(\text{No shared birthday})=\frac{365}{365}\times\frac{364}{365}\times\frac{363}{365}\times\cdots\times\frac{336}{365}$

2. Use the formula for the probability of no shared birthdays:

$P(\text{No shared birthday})=\frac{365!}{(365-30)!\times 365^{30}}\approx 0.294$

3. Subtract from 1 to find the probability of at least two people sharing a birthday:

$P(\text{At least one shared birthday})=1-0.294\approx 0.706$

11. Calculate the median of the following data set: 7, 3, 5, 9, 2, 8, 4. 1 / 1 point

5

✓ Nice work

The median is 5.

1. Order the data set in ascending order:

2, 3, 4, 5, 7, 8, 9

2. Identify the middle value:

Since there are seven data points, the median is the 4th value: 5

12. A class of 20 students has the following test scores: 45, 56, 78, 90, 65, 88, 70, 50, 92, 84, 72, 68, 81, 60, 75, 85, 87, 93, 77, 55. Calculate the arithmetic mean of the test scores. 0 / 1 point

73.55

⊗ Try again

The arithmetic mean is 73.4.

1. Sum the test scores:

$$45+56+78+90+65+88+70+50+92+84+72+68+81+60+75+\\85+87+93+77+55=1471$$

2. Divide by the number of students:

$\frac{1471}{20}=73.43\approx 73.4$

13. Determine the range, mean, and median of the following data set: 15, 22, 27, 10, 19, 24, 18, 16, 12, 20.

Please insert your answers without any spaces. So, if the range is 10, the mean is 19 and the median is 14, insert the answer as

10,19,14 1 / 1 point

17,18.3,18.5

✓ Nice work

Default Feedback:

Range: 17

Mean: 18.3

Median: 18.5

1. Order the data set in ascending order:

10, 12, 15, 16, 18, 19, 20, 22, 24, 27

2. Calculate the range:

$27-10=17$

3. Calculate the mean:

$\frac{10+12+15+16+18+19+20+22+24+27}{10}=\frac{183}{10}=18.3$

4. Identify the median:

Since there are 10 data points, the median is the average of the 5th and 6th values: $\frac{18+19}{2}=18.5$

14. A teacher wants to calculate the standard deviation of the following test scores: 50, 60, 70, 80, 90. Determine the standard deviation. 1 / 1 point

14.14

✓ Nice work

The standard deviation is 14.14.

1. Calculate the mean:

$\text{Mean}=\frac{50+60+70+80+90}{5}=70$

2. Subtract the mean from each score and square the result:

$(50-70)^2=400$

$(60-70)^2=100$

$(70-70)^2=0$

$(80-70)^2=100$

$(90-70)^2=400$

3. Sum the squared differences:

$400+100+0+100+400=1000$

4. Divide by the number of observations minus one (n-1) for sample variance:

$\frac{1000}{5}=200$

5. The variance is 200.

6. Take the square root of the variance for the standard deviation:

$\sqrt{200}\approx 14.14$

15. A student records the following number of hours studied each day for a week: 2, 3, 5, 2, 4, 3, 5. Calculate the median number of hours studied. 1 / 1 point

3

✓ Nice work

The median is 3.

1. Order the data set in ascending order:

2, 2, 3, 3, 4, 5, 5

2. Identify the middle value:

Since there are seven data points, the median is the 4th value: 3